

CS-FLEURS: A Massively Multilingual and Code-Switched Speech Dataset

Brian Yan¹, Injy Hamed², Shuichiro Shimizu³, Vasista Lodagala⁴, William Chen¹, Olga Iakovenko⁵, Bashar Talafha⁶, Amir Hussein⁷, Alexander Polok⁸, Calvin Chang¹, Dominik Klement⁸, Sara Althubaiti⁴, Puyuan Peng⁹, Matthew Wiesner⁷, Tamar Solorio², Ahmed Ali⁴, Sanjeev Khudanpur⁷, Shinji Watanabe¹

¹Carnegie Mellon University, ²Mohamed bin Zayed University of Artificial Intelligence, ³Kyoto University, ⁴Humain, ⁵University of Sheffield, ⁶University of British Columbia, ⁷Johns Hopkins University, ⁸Brno University of Technology, ⁹University of Texas at Austin

Abstract

We present CS-FLEURS, a new dataset for developing and evaluating code-switched speech recognition and translation systems beyond high-resourced languages. CS-FLEURS consists of 4 test sets which cover in total 113 unique code-switched language pairs across 52 languages: 1) a 14 X-English language pair set with real voices reading synthetically generated code-switched sentences, 2) a 16 X-English language pair set with generative text-to-speech 3) a 60 {Arabic, Mandarin, Hindi, Spanish}-X language pair set with the generative text-to-speech, and 4) a 45 X-English lower-resourced language pair test set with concatenative text-to-speech. Besides the four test sets, CS-FLEURS also provides a training set with 128 hours of generative text-to-speech data across 16 X-English language pairs. Our hope is that CS-FLEURS helps to broaden the scope of future code-switched speech research.

Index Terms: code-switching, code-switched speech recognition, multilingual speech recognition and translation

1. Introduction

In the current era of massively multilingual speech processing, practitioners are developing and evaluating systems on hundreds of languages [1, 2, 3, 4, 5]. This increasingly global impact is due in large part to dataset construction efforts, which have enabled pre-training on raw untranscribed speech [6, 7] and supervised training on transcribed or pseudo-labeled speech [8, 9, 10, 11], as well as broad benchmarking on standardized evaluation sets across many languages [12, 13]. However, these data resources are collections of predominately *monolingual* utterances - we currently lack broad *code-switched* speech data, where utterances consist of more than one language.

Despite its prevalence in multilingual communities, conversational code-switched speech is particularly challenging to collect at scale. Unlike monolingual data, which can be sourced from a wide range of speakers, code-switched speech requires bilingual or multilingual speakers who naturally mix languages in conversation [14]. Additionally, code-switching tends to occur sporadically and unpredictably, making it difficult to capture in large, curated datasets. Even when data is available, standardizing it across language pairs presents another challenge. These factors make it difficult to create a single, broadly representative dataset of code-switched speech; instead, practitioners have focused on constructing dedicated corpora for their particular communities of interest [15, 16, 17, 18, 19, 20].

In this work, we take an alternative approach: rather than relying on the collection of natural conversational code-

Additional Contributing Authors: Chih-Chen Chen, Zhen Wu, Karim Benharrak, Anuj Diwan, Samuele Cornell, Eunjung Yeo, Kwanghee Choi, Carlos Carvalho, Karen Rosero

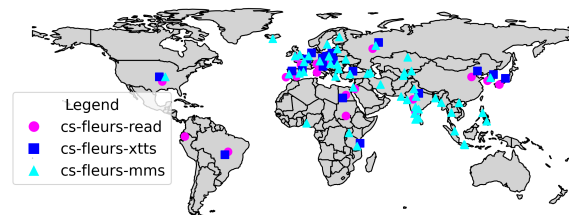


Figure 1: Illustration of where the 52 languages covered by CS-FLEURS are spoken around the world.

switched speech, we use a combination of read speech and synthetic speech over synthetically generated code-switched text. Although our approach does not capture code-switched speech that occurs in conversation, we are able to use (1) **standard code-switching patterns**, (2) a **standard textual domain**, and (3) a **standard audio domain** across language pairs. By controlling for these naturally occurring variations when constructing the dataset, we can then ask: *how do models perform on code-switched speech across different language pairs?*

We present CS-FLEURS: a massively multilingual and code-switched ASR and ST dataset consisting of 52 languages and 113 unique code-switched pairs across three subsets:

- CS-FLEURS-READ: 14 X-English pairs, read speech
- CS-FLEURS-XTTS: 76 pairs across 17 langs, generative TTS
- CS-FLEURS-MMS: 45 X-English pairs, concatenative TTS

To the best of our knowledge, this is the broadest single collection of code-switched speech data both in terms of languages covered and number of unique code-switched pairs.

This dataset supports model benchmarking across 113 diverse pairs and model training across 16 X-English pairs. Figure 1 illustrates the language coverage and Table 1 summarizes the dataset with comparison to other code-switched corpora. In addition to the dataset, this work also describes several empirical findings. First, we show that Whisper [2] struggles with transcribing code-switched speech, especially for language pairs with distinct scripts. However, we also show that Whisper is relatively robust at translating code-switched speech to English, suggesting that code-switched ST is an easier task than ASR. Finally, we show that synthesizing code-switched speech data is an effective form of training data augmentation, improving model performance on both seen and unseen language pairs.

2. Dataset

CS-FLEURS builds upon a line of prior works. The FLoRes-101 dataset [21] consists of 3001 sentences from English Wikipedia translated in 101 languages by human translators.

Table 1: CS-FLEURS vs other code-switched speech corpora.

Dataset	Langs CS-Pairs		Domain	Speech Type	Hours
ArzEn [20]	2	1	Conversation	Spontaneous	12
Bangor Miami [15]	2	1	Conversation	Spontaneous	35
SEAME [16]	2	1	Conversation	Spontaneous	192
ESCWA [18]	3	2	Conversation	Spontaneous	3
MUCS [19]	3	2	Lecture	Structured	160
Soapiers [17]	5	4	TV	Scripted	14
CS-FLEURS	52	113	Wikipedia	Read/Synthetic	294

Table 2: The 113 code-switched language pairs covered.

CS-Pairs	Matrix	Embedded	Read XTTS MMS		
7	ara deu fra hin por rus spa	eng	✓	✓	✓
5	ces cmn ita jpn kor	eng	✓	✓	
2	slk tel	eng	✓		
3	nld pol tur	eng		✓	✓
1	hun	eng		✓	
60	ara cmn hin spa	ara fra deu ita por pol tur rus nld ces spa cmn jpn hun kor hin		✓	
35	ben bul cat ceb cym ell fin guj heb hun ind isl jav kan kaz kir lav lug mal mar mya pan ron swe swl tam tel tgl tha ukr urd uzb yor zlm	eng			✓

Then, the FLEURS dataset [13] took 2009 of those sentences from FLoReS-101 and recorded read speech in 102 languages with three speakers per language. Now, CS-FLEURS takes the 2009 utterances from FLEURS to generate code-switched text and speech across 113 language pairs, keeping the same train/dev/test splits from FLEURS.

We follow the Matrix Language-Frame model [22]; for each code-switched pair, one language, referred to as Matrix, provides grammatical structure while a second language, referred to as Embedded, provides words or morphological units that are inserted within the sentence. We refer to language pairs as Matrix-Embedded; for instance, Mandarin-English refers to a Mandarin matrix sentence with English words embedded. Table 2 shows a breakdown of the code-switched language pairs covered by CS-FLEURS across the CS-FLEURS-READ, CS-FLEURS-XTTS, and CS-FLEURS-MMS subsets. Table 3 provides additional summary statistics.

2.1. Collecting Read Speech Across 14 Language Pairs

CS-FLEURS-READ consists of read speech from bilingual speakers across 14 X-English language pairs (1-4 speakers per language pair, 2:1 M:F ratio overall). CS-FLEURS-READ is generated using the test set of FLEURS and is intended for model benchmarking. As such, this set has been human validated. In terms of language coverage, the limiting factor for constructing this set was the availability of bilingual readers. Finally, in this set we regard the 14 non-English languages as Matrix and English as Embedded.

2.1.1. Generating accurate and fluent code-switched text

We use a large language model (LLM) backbone for generating code-switched text for CS-FLEURS-READ. Specifically, by utilizing X-English parallel monolingual sentences from FLEURS test set, we prompt GPT-4o [23] to generate the equivalent code-

Table 3: Summary statistics for CS-FLEURS.

Statistic	READ	XTTS				MMS
	Test	Train	Dev	Test1	Test2	Test
Duration (hours)	17	128	15	36	42	56
Tokens (words)	128k	889k	105k	257k	300k	315k
Matrix Langs	14	16	16	16	4	45
Embedded Langs	1	1	1	1	15	1
Total CS Pairs	14	16	16	16	60	45
Same-Script Pairs	7	10	10	10	9	22
Distinct-Script Pairs	7	6	6	6	51	23

switched sentences in three different manners:

- *GPT-Base*: feeding only the paired monolingual sentences
- *GPT-EC*: feeding paired monolingual sentences and a set of valid English words to be embedded, determined by the Equivalence Constraint theory [24] so as not to violate the syntactic rules of both languages, as implemented in [25]
- *GPT-Pred*: feeding paired monolingual sentences and a set of valid English words to be embedded, determined by a predictive model [26], which given an English sentence, predicts the plausible English words to be embedded in a corresponding code-switched sentence

Table 4 shows an example of a full prompt for a given sentence pair. *GPT-EC* and *GPT-Pred* prompts include English keywords for embedding into the matrix language. Further, to encourage morphological code-switching, where more than one language occurs *within the same word*, we prompt GPT-4o with an in-context example, as illustrated in the table. 10 out of 14 languages were prompted as such.¹

The resultant generations are not guaranteed to be (1) code-switched, (2) accurate, preserving the meaning of the original monolingual sentences, or (3) fluent, plausibly produced by a human. We ask our bilingual readers to reject sentences which are not code-switched or not accurate. Sentences are not rejected on fluency, but we collect ratings from 0-2, with 0 indicating unnaturalness, 1 indicating somewhat natural, and 2 indicating perfect naturalness.

As shown in Table 5, the reject rate was fairly consistent across the different prompts. In terms of fluency, *GPT-Pred* scored highest while *GPT-EC* was lowest; this result correlates with the frequency of switching, as measured by the Code-Mixing Index (CMI) [27]. We also noted that *GPT-EC* embeds function words a high rate, which bilingual readers found to be rather unnatural. *GPT-Pred* on the other hand primarily embeds content words, which resulted in over 80% of generations to be rated somewhat or perfectly natural.

2.1.2. Reading and recording code-switched speech

To facilitate the collection of read speech, we developed an open-source toolkit.² We distributed generated code-switched text to 21 bilingual readers. For each sentence, we randomly selected one of the three aforementioned prompts, standardizing the selection across all language pairs. The bilingual readers were provided with the monolingual references along with the code-switched sentences. They were instructed to 1) verify that the sentence is indeed code-switched, 2) the sentence has

¹ 10 w/ morphological example: ara, ces, deu, ita, kor, por, rus, slk, spa, tel, tam; 4 w/o: cmn, fra, hin, jpn

² <https://github.com/brianyan918/sentence-recorder/tree/codeswitching>

Table 4: The prompt consists of three main parts: (1) task description and the paired monolingual sentences to be synthesized (outlined in black); (2) the set of valid English words in the case of GPT-EC and GPT-Pred (outlined in blue); and (3) morphological code-switching example in the case of languages where this type of switching is common (outlined in purple).

You are a bilingual Spanish-English speaker, you will help translate Spanish and English sentences into a code-mixed sentence. The code-mixed sentence should contain words from both languages, each written in the correct language script. We will provide you with both Spanish and English sentences in addition to some English keywords that need to be present in the produced code-mixed sentence.

You need to produce morphological code-switching when appropriate to achieve better fluency. The following is an example:

Spanish: Mi amigo se enojó cuando le revelaron el final de la película.
English: My friend freaked out when someone spoiled the end of the movie. English keywords: freak out, spoiled

A fluent code-mixed sentence would be: Mi amigo se fricó cuando le spoilieron el final de la película.

Please produce the code-mixed sentence for the following:
Spanish sentence: {Spanish_sentence}
English sentence: {English_sentence}
English keywords: {English_keywords}

Only provide the code-mixed sentence without explanation or extra text.

Table 5: Human validation on LLM-generated code-switched text across 14 X-English pairs. CMI=Code-Mixing Index.

Prompt	Reject(%)	Fluency					CMI
		0(%)	1(%)	2(%)	Avg		
GPT-Base	8.54	31.66	25.44	34.36	1.03	25.50	
GPT-EC	12.15	44.71	21.22	21.92	0.74	30.98	
GPT-Pred	11.76	19.26	25.25	43.73	1.28	21.05	

the same general meaning as the monolingual reference, 3) read and record the sentence, and 4) rate the fluency of the sentence. Recordings were then validated by a human listener.

2.2. Synthesizing Speech Across 111 Language Pairs

Next, we describe two synthetic speech sets designed to complement CS-FLEURS-READ by covering additional language pairs and providing code-switched training data.

CS-FLEURS-XTTS consists of synthetic speech from the XTTS-v2 model [28], which supports 17 languages, across 16 X-English language pairs as well as 60 non-English language pairs (15 Arabic-X, 15 Hindi-X, 15 Mandarin-X, 15 Spanish-X). For the 16 X-English language pairs, we generate train, dev, and test sets, regarding the 16 non-English languages as matrix and English as embedded — 12 of these language pairs overlap with CS-FLEURS-READ. The 60 non-English language pairs are test-only and we regard Arabic, Hindi, Mandarin, or Spanish as the matrix and the other language as embedded — none of these language pairs overlap with the other two subsets.

CS-FLEURS-MMS consists of synthetic speech from MMS TTS models [5] across 45 X-English language pairs, many of which are lower-resourced — only 10 of these language pairs overlap with the other two subsets. This set is test-only and we regard the non-English languages as matrix.

Both sets are not human validated due to lack of bilingual speakers. Instead, we use language-universal forced alignment to filter low quality generations.

2.2.1. Cheaply and flexibly generating code-switched text

While the LLM backbone described in §2.1.1 can generate natural and morphologically rich code-switching, there are also

Table 6: Comparison of the code-switching frequency, as measured by Code-Mixing Index (CMI), of text generated by LLM prompts and align-then-swap (Swap) across 12 X-English language pairs. M=Mean; SD=Standard Deviation.

Type	ara	ces	cmn	deu	fra	hin	ita	jpn	kor	por	rus	spa	M	SD
LLM	12.3	30.4	24.1	27.8	36.0	18.5	37.0	18.5	7.0	37.2	17.7	39.8	25.5	10.8
Swap	20.4	20.1	14.11	16.8	15.0	13.6	16.8	11.2	25.9	17.0	20.6	15.5	17.2	4.0

a number of drawbacks that limit scalability aside from the high computational (or API) cost. The LLM-based methods required human validation to rejected around 10% of the sentences across 14 X-English language pairs; it is reasonable to expect a higher reject rate for lower resourced languages.

Further, we found that the LLM-based methods produced a wide range of code-switching frequency, as measured by CMI [27], across language pairs. This variance is shown in the first line of Table 6: the standard deviation of the CMI across 12 language pairs in the human validated CS-FLEURS-READ is 10.8. We therefore opt for a more simple and rigid method for generating the code-switched text in CS-FLEURS-XTTS and CS-FLEURS-MMS, referred to as align-then-swap.

The align-then-swap procedure is simple. We first obtain word-level alignment using AwesomeAlign [29], a 104 language mBERT model [30] fine-tuned towards word alignment objectives using parallel text across five language pairs. Notably, AwesomeAlign generalizes to other languages that were part of the mBERT pre-training but unseen during the word alignment fine-tuning stage [29]. Next we randomly select 30% nouns, verbs, abverbs, and adjectives, tagged using Stanza [31], to be swapped with the aligned embedded language words.

Additionally, we take care of one-to-many word alignments (matrix-to-embedded) by inserting words in the order that they originally appeared in the monolingual embedded language sentence. We also take care of character-based languages (Mandarin and Japanese) by first re-segmenting to words using Stanza [31]; Korean text in FLEURS was already word segmented, so this extra step was not necessary.

As shown in Table 6, the resultant CMI from align-then-swap (Swap) is more consistent across language pairs than its LLM-based counterpart: the standard deviation is only 4.0, which is 60% lower. This simpler, yet more consistent, method of generating code-switched text is therefore preferable for the long tail of lower resourced languages and rare language pairs.

2.2.2. Generative code-switched synthesis with XTTS

XTTS-v2 [28] is a multilingually trained TTS model based on a GPT-2 encoder that predicts VQ-VAE units which then latently condition a HiFi-GAN decoder. The model was trained on 17 languages, which are distinguished by appending a language ID token (e.g. [en]) at the start of textual inputs. XTTS-v2 also incorporates a voice conversion component by conditioning both the GPT-2 encoder and VQ-VAE decoder on speaker information; in practice only single utterance from the target speaker is necessary to enable voice conversion.

To synthesize code-switched speech using XTTS-v2, we feed generated code-switched text to the model with the matrix language ID token. Mandarin, Japanese, and Korean text is romanized, in the same manner as the original XTTS-v2 training. We use a matrix language speaker from the FLEURS dataset for the voice conversion component. XTTS-v2 produces 24khz speech, which we then down-sample to 16khz.

Table 7: Comparison of the filtered vs accepted synthetic speech. Filtering was done by forced alignment score (FAS).

Subset	XTTS-TEST1				MMS-TEST			
	FAS↑	CER↓	UTMOS↑	SCD↓	FAS↑	CER↓	UTMOS↑	SCD↓
Filtered	-0.68	32.95	2.46	1.12	-0.96	43.10	3.02	1.90
Accepted	-0.26	18.46	2.64	1.07	-0.47	40.07	3.04	1.87

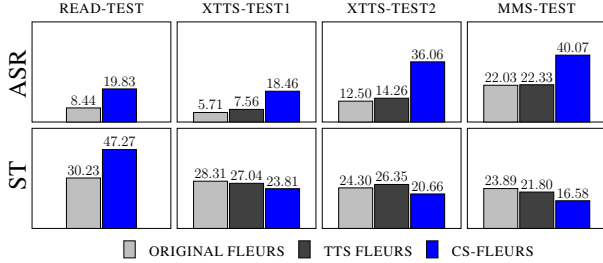


Figure 2: Whisper-Large-v3 ASR (CER↓) and ST to English (BLEU↑) on CS-FLEURS test sets (blue). Performance on two monolingual control sets are shown for comparison (gray).

2.2.3. Concatenative code-switched synthesis with MMS-TTS

To cover additional language pairs, we use a concatenative approach that generates monolingual TTS in segments using MMS-TTS [5]. We concatenate the monolingual segments, inserting 100ms of silence in between. Since MMS-TTS is a single-speaker VITS style model, the resulting concatenated code-switched speech contains unnatural artifacts and speaker changes; however the content is correct. Unlike the XTTS-v2 outputs, MMS-TTS outputs do not contain any accented speech.

2.2.4. Universal forced alignment filtering

To perform quality control, we use a universal forced alignment model, MMS-ZS [32], which was trained to align speech in thousands of languages to romanized text - this model provides a language-universal measure of intelligibility. Utterances with the 5% lowest length-normalized forced alignment score (FAS) are filtered within each language pair.

In Table 7, we report the average FAS of the filtered vs accepted portions of XTTS-TEST1 and MMS-TEST, along with Whisper-Large-v3 [2] character error rate (CER), the UTMOS (1-5) [33] automatic naturalness metric, and the number of speakers changes identified by the Pyannote-3.1 [34] diarization model (SCD). For XTTS-TEST1, filtering has a large effect on Whisper CER and a moderate effect on UTMOS, indicating that the generative synthesis method is prone to producing unintelligible and unnatural speech. On the other hand for MMS-TEST, filtering has less effect on CER and UTMOS, indicating that concatenative synthesis is less error prone; however, SCD shows that this method results in frequent speaker changes.

3. Experiments

In this section, we describe several empirical findings based on CS-FLEURS. For ASR, we measure case insensitive and unpunctuated character error rate (CER). For ST (to English), we measure case insensitive and unpunctuated BLEU [35].

3.1. Benchmarking

How does Whisper perform on code-switched speech as compared to monolingual speech? To answer this, we compare ASR and ST (to English) performance on each CS-FLEURS

Table 8: Comparison of Whisper-Large-v3 ASR (CER↓) performance on same-script vs distinct-script language pairs.

CS Pair Type	READ Test	XTTS		MMS Test
		Test1	Test2	
Same Script	7.32	8.49	9.92	28.26
Distinct Script	32.33	37.17	40.67	51.62

Table 9: Results on the effect of training data augmentation.

Data Augment	FLEURS		CS-FLEURS-READ	
	12 Seen	2 Unseen	12 Seen	2 Unseen
✓	14.38	8.93	31.77	29.62
	12.67	8.51	26.24	27.77

test set with two monolingual control sets consisting of monolingual speech from all matrix languages in each respective CS-FLEURS set: 1) the original FLEURS and 2) a TTS version of FLEURS, which is monolingual. The latter, referred to as TTS FLEURS, allows us to compare results on TTS-based code-switched speech with the matching TTS-based monolingual speech, eliminating the possibility that Whisper simply is not robust to XTTS or MMS-TTS speech.

As shown in Figure 2, ASR CER is over 2x higher on CS-FLEURS than the original monolingual FLEURS - this degradation is largest on the non-English language pair set (XTTS-TEST2). However, the discrepancy between ST performance on monolingual and code-switched speech is low for XTTS and MMS sets. Further, Whisper actually performs better on READ-TEST than the original FLEURS - this can be attributed to the high rate of X-English code-switching (see Table 6). These results indicate that **directly translating code-switched speech is less erroneous than relying on transcription**, although we leave benchmarking ST for non-English targets to future work.

How does Whisper perform on code-switched speech across different language pairs? As shown in Table 8, performance on distinct-script language pairs is significantly worse than same-script language pairs - on average CER is 3x higher. These results suggest that **code-switched ASR performance is limited by the ability to interchangeably produce two distinct scripts** within a single utterance-level decoding.

3.2. Training

Finally, we investigate the question: *how useful is synthetic data for training models?* To answer this, we train two models using the ESPnet toolkit [36]: one using the original FLEURS training data and a second using the original FLEURS plus CS-FLEURS (XTTS-TRAIN). Both models are trained for the same number of iterations following the self-conditioned XLSR-based recipe described in [37]. We then evaluated on READ-TEST, which consists of 14 language pairs; 12 of these are seen in XTTS-TRAIN and 2 are unseen. We also report the monolingual FLEURS result for the matrix languages of these 12 seen and 2 unseen language pairs. As shown in Table 9, **training on synthetic code-switched data improved performance on both seen and unseen language pairs**.

4. Conclusion

CS-FLEURS is a hybrid read/synthetic speech dataset for developing and evaluating code-switched systems across 52 languages and 113 unique code-switched language pairs. By using this dataset, which controls for in code-switching patterns, text domain, and audio domain, we find that transcribing speech across distinct-script language pairs remains difficult.

5. References

- [1] X. Li, F. Metze, D. R. Mortensen, A. W. Black, and S. Watanabe, "Asr2k: Speech recognition for around 2000 languages without audio," *Proc. Interspeech 2022*, pp. 4885–4889, 2022.
- [2] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc ICML*. PMLR, 2023, pp. 28 492–28 518.
- [3] Y. Zhang, W. Han, J. Qin, Y. Wang, A. Bapna, Z. Chen, N. Chen, B. Li, V. Axelrod, G. Wang *et al.*, "Google usm: Scaling automatic speech recognition beyond 100 languages," *arXiv preprint arXiv:2303.01037*, 2023.
- [4] Y. Peng, J. Tian, W. Chen, S. Arora, B. Yan, Y. Sudo, M. Shakeel, K. Choi, J. Shi, X. Chang, J. weon Jung, and S. Watanabe, "Owsm v3.1: Better and faster open whisper-style speech models based on e-branchformer," in *Interspeech 2024*, 2024, pp. 352–356.
- [5] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Kundu, A. Elkahky, Z. Ni, A. Vyas, M. Fazel-Zarandi *et al.*, "Scaling speech technology to 1,000+ languages," *Journal of Machine Learning Research*, vol. 25, no. 97, pp. 1–52, 2024.
- [6] J. Kahn, M. Riviere, W. Zheng, E. Kharitonov, Q. Xu, P.-E. Mazaré, J. Karadayi, V. Liptchinsky, R. Collobert, C. Fuegen *et al.*, "Libri-light: A benchmark for asr with limited or no supervision," in *Proc ICASSP*. IEEE, 2020, pp. 7669–7673.
- [7] C. Wang, M. Riviere, A. Lee, A. Wu, C. Talnikar, D. Haziza, M. Williamson, J. Pino, and E. Dupoux, "Voxpopuli: A large-scale multilingual speech corpus for representation learning, semi-supervised learning and interpretation," in *Proc ACL-IJCNLP*, 2021.
- [8] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: an asr corpus based on public domain audio books," in *Proc ICASSP*. IEEE, 2015, pp. 5206–5210.
- [9] V. Pratap, Q. Xu, A. Sriram, G. Synnaeve, and R. Collobert, "Mls: A large-scale multilingual dataset for speech research," in *Interspeech 2020*, 2020, pp. 2757–2761.
- [10] R. Ardila, M. Branson, K. Davis, M. Kohler, J. Meyer, M. Henretty, R. Morais, L. Saunders, F. Tyers, and G. Weber, "Common voice: A massively-multilingual speech corpus," in *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 2020, pp. 4218–4222.
- [11] X. Li, S. Takamichi, T. Saeki, W. Chen, S. Shiota, and S. Watanabe, "Yodas: Youtube-oriented dataset for audio and speech," in *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2023, pp. 1–8.
- [12] J. Shi, D. Berrebbi, W. Chen, E.-P. Hu, W.-P. Huang, H.-L. Chung, X. Chang, S.-W. Li, A. Mohamed, H. yi Lee, and S. Watanabe, "Ml-superb: Multilingual speech universal performance benchmark," in *Interspeech 2023*, 2023, pp. 884–888.
- [13] A. Conneau, M. Ma, S. Khanuja, Y. Zhang, V. Axelrod, S. Dalmia, J. Riesa, C. Rivera, and A. Bapna, "Fleurs: Few-shot learning evaluation of universal representations of speech," in *2022 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2023.
- [14] P. Auer, *Code-switching in conversation: Language, interaction and identity*. Routledge, 2013.
- [15] M. Deuchar, P. Davies, J. Herring, M. C. P. Couto, and D. Carter, "Building bilingual corpora," *Advances in the Study of Bilingualism*, pp. 93–111, 2014.
- [16] D.-C. Lyu, T. P. Tan, E. Chng, and H. Li, "Seame: a mandarin-english code-switching speech corpus in south-east asia," in *Interspeech*, vol. 10, 2010, pp. 1986–1989.
- [17] E. van der Westhuizen and T. Niesler, "A first South African corpus of multilingual code-switched soap opera speech," in *Proc LREC*, 2018.
- [18] S. A. Chowdhury, A. Hussein, A. Abdelali, and A. Ali, "Towards one model to rule all: Multilingual strategy for dialectal code-switching arabic asr," in *Interspeech 2021*, 2021, pp. 2466–2470.
- [19] A. Diwan, R. Vaideeswaran, S. Shah, A. Singh, S. Raghavan, S. Khare, V. Unni, S. Vyas, A. Rajpuria, C. Yarra, A. Mittal, P. K. Ghosh, P. Jyothi, K. Bali, V. Seshadri, S. Sitaram, S. Bharadwaj, J. Nanavati, R. Nanavati, and K. Sankaranarayanan, "Mucs 2021: Multilingual and code-switching asr challenges for low resource indian languages," in *Interspeech 2021*, 2021, pp. 2446–2450.
- [20] I. Hamed, N. Habash, S. Abdennadher, and N. T. Vu, "ArEn-ST: A three-way speech translation corpus for code-switched Egyptian Arabic-English," in *Proc WANLP*, 2022, pp. 119–130.
- [21] N. Goyal, C. Gao, V. Chaudhary, P.-J. Chen, G. Wenzek, D. Ju, S. Krishnan, M. Ranzato, F. Guzmán, and A. Fan, "The flores-101 evaluation benchmark for low-resource and multilingual machine translation," *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 522–538, 2022.
- [22] C. Myers-Scotton, *Duelling languages: Grammatical structure in codeswitching*. Oxford University Press, 1997.
- [23] A. Hurst, A. Lerer, A. P. Goucher, A. Perelman, A. Ramesh, A. Clark, A. Ostrow, A. Welihinda, A. Hayes, A. Radford *et al.*, "Gpt-4o system card," *arXiv preprint arXiv:2410.21276*, 2024.
- [24] S. Poplack, "Sometimes i'll start a sentence in Spanish y termino en Español: Toward a typology of code-switching," *The bilingualism reader*, vol. 18, no. 2, pp. 221–256, 1980.
- [25] G. Kuwanto, C. Agarwal, G. I. Winata, and D. T. Wijaya, "Linguistics theory meets llm: Code-switched text generation via equivalence constrained large language models," *arXiv preprint arXiv:2410.22660*, 2024.
- [26] I. Hamed, N. Habash, S. Abdennadher, and N. T. Vu, "Investigating lexical replacements for arabic-english code-switched data augmentation," in *Proc LoResMT*, 2023, pp. 86–100.
- [27] B. Gambäck and A. Das, "Comparing the level of code-switching in corpora," in *Proceedings of the 10th International Conference on Language Resources and Evaluation (LREC)*, Reykjavik, Iceland, 2014, pp. 1850–1855.
- [28] E. Casanova, K. Davis, E. Gölgel, G. Gökner, I. Gulea, L. Hart, A. Aljafari, J. Meyer, R. Morais, S. Olayemi *et al.*, "Xtts: a massively multilingual zero-shot text-to-speech model," *arXiv preprint arXiv:2406.04904*, 2024.
- [29] Z.-Y. Dou and G. Neubig, "Word alignment by fine-tuning embeddings on parallel corpora," in *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, 2021, pp. 2112–2128.
- [30] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings NAACL*, J. Burstein, C. Doran, and T. Solorio, Eds. Association for Computational Linguistics, 2019, pp. 4171–4186.
- [31] P. Qi, Y. Zhang, Y. Zhang, J. Bolton, and C. D. Manning, "Stanza: A python natural language processing toolkit for many human languages," *arXiv preprint arXiv:2003.07082*, 2020.
- [32] J. Zhao, V. Pratap, and M. Auli, "Scaling a simple approach to zero-shot speech recognition," *arXiv preprint arXiv:2407.17852*, 2024.
- [33] T. Saeki, D. Xin, W. Nakata, T. Koriyama, S. Takamichi, and H. Saruwatari, "Utmos: Uto-kyo-sarulab system for voicemos challenge 2022," in *Interspeech 2022*, 2022, pp. 4521–4525.
- [34] H. Bredin, "pyannote.audio 2.1 speaker diarization pipeline: principle, benchmark, and recipe," in *Proc. INTERSPEECH 2023*, 2023.
- [35] M. Post, "A call for clarity in reporting BLEU scores," in *Proceedings of the Third Conference on Machine Translation: Research Papers*, 2018.
- [36] S. Watanabe, T. Hori, S. Karita, T. Hayashi, J. Nishitoba, Y. Unno, N. Enrique Yalta Soplin, J. Heymann, M. Wiesner, N. Chen, A. Renduchintala, and T. Ochiai, "Espnet: End-to-end speech processing toolkit," in *Interspeech 2018*, 2018, pp. 2207–2211.
- [37] W. Chen, B. Yan, J. Shi, Y. Peng, S. Maiti, and S. Watanabe, "Improving massively multilingual asr with auxiliary ctc objectives," in *Proc ICASSP*. IEEE, 2023, pp. 1–5.